Analysis of Scientific Data – Semester 1, 2026

Applied Class 5 – Solutions

Part A – Normal Distribution

Suppose that pulse rates while completing the student survey come from a Normal distribution with mean 71.7 bpm and standard deviation 11.7 bpm.

a) What is the probability that a random student has a pulse rate of at least 90 bpm?

$$P(X \geq 90) = P\left(\frac{X - 71.7}{11.7} \geq \frac{90 - 71.7}{11.7}\right) = P(Z \geq 1.564) = 1 - P(Z \leq 1.564)$$

In R we can calculate $P(Z \leq 1.564)$ as

```
pnorm(1.564)
[1] 0.9410912
```

So $P(Z \geq 1.564) = 1 - 0.9411 = 0.0589$.

b) What is the probability that a random student has pulse rate between 60 and 80 bpm?

$$P(60 \leq X \leq 80) = P\left(\frac{60 - 71.7}{11.7} \leq \frac{X - 71.7}{11.7} \leq \frac{80 - 71.7}{11.7}\right) = P(-1 \leq Z \leq 0.7094)$$

$$= P(Z \leq 0.7094) - P(Z \leq -1) = 0.7610 - 0.1587 = 0.6023$$

c) What value would put a student in the bottom 10% of pulse rates?

In this question we want to find the value q (quantile) such that

$$0.10 = P(X \leq q).$$

Again, we standardize X

$$0.1 = P\left(\frac{X - 71.7}{11.7} \leq \frac{q - 71.7}{11.7}\right) = P\left(Z \leq \frac{q - 71.7}{11.7}\right)$$

Using the qnorm function in R

```
qnorm(0.1)
[1] -1.281552
```

That is, $P(Z \leq -1.2816) = 0.1$. So we need to solve the equation

$$-1.2816 = \frac{q - 71.7}{11.7}$$

$$-1.2816 \times 11.7 = q - 71.7$$

$$q = 71.7 - 1.2816 \times 11.7 = 56.706$$

So a pulse rate of 56.706 bpm would put a student in the bottom 10% of pulse rates.

d) In a random sample of 5 students, what is the probability that at least 3 of them have a pulse rate over 90 bpm while completing the survey?

From part (a) we know the probability that a student has a pulse rate greater than 90 bpm is 0.0589. Let X be the random variable representing the number of students having a pulse rate over 90 bpm in a sample of 5 students. Assuming students are selected independently at random from the student population, X has a Binomial(5, 0.0589) distribution.

We want $P(X \geq 3)$ which can be evaluated in R as

```
sum(dbinom(x=3:5,size=5,prob=0.0589)
[1] 0.001867087
```

Part B – Paired T Test

Alice recruited nine pairs of identical twins for a study of two cholesterol-reducing drugs, A and B, with the aim of showing that drug A gave higher reductions in cholesterol than drug B. One of the twins in each pair was given drug A and the other was given drug B, where the choice was made at random. The amount by which cholesterol was reduced in each subject (mg/dL) is given in the following table:

Pair	1	2	3	4	5	6	7	8	9
Drug A	74	55	61	47	53	74	52	40	50
Drug B	63	58	49	41	50	69	59	31	44
Difference	11	-3	12	6	3	5	-7	9	6

The sample mean difference in cholesterol reduction between drug A and drug B was 4.67 mg/dL with a sample standard deviation of 6.265 mg/dL.

a) State the null and alternative hypotheses of interest in terms of μ, the mean difference in cholesterol reduction between drug A and drug B in twins in this population.

H_0: $\mu = 0$

H_1: $\mu > 0$

b) Calculate the t statistic to test this null hypothesis.

$$t_8 = \frac{4.67 - 0}{6.265/\sqrt{9}} = 2.236$$

i.e. 4.67 mg/dL is 2.236 standard errors above the hypothesised difference of 0 mg/dL.

c) What is the corresponding p-value? What do you conclude?

P-value is $P(\bar{X} \geq 4.67) = P(T_8 \geq 2.236) = 0.0279$ using `1-pt(2.2362,df=8)`

This is **moderate** evidence against H_0, suggesting drug A does give a higher reduction in cholesterol than drug B.

d) Based on this data, calculate a 95% confidence interval for the mean difference in cholesterol reduction between drug A and drug B.

The critical value for 95% confidence is qt(.975, df=8) = 2.306. Thus 95% CI is

$$4.67 \pm 2.306 \times \frac{6.265}{\sqrt{9}} = 4.67 \pm 4.82 \text{ mg/dL}$$

This gives the interval $(-0.15, 9.49)$ mg/dL. That is, we are 95% confident that the reduction with drug A is between 0.15 mg/dL lower and 9.49 mg/dL higher than drug B.

You may find the following output from R useful in answering the questions on this sheet.

```
> sum(dbinom(x=3:5,size=5,prob=.0589))
[1] 0.001867087
> sum(dbinom(x=3:5,size=5,prob=.1142))
[1] 0.01245883
> pnorm(-2.114)
[1] 0.01725763
> pnorm(1.564)
[1] 0.9410912
> pnorm(-1.564)
[1] 0.05890878
> pnorm(0.7094017)
[1] 0.7609624
> pnorm(-1)
[1] 0.1586553
> qnorm(.9)
[1] 1.281552
> qnorm(.1)
[1] -1.281552
```

```
> pt(0.7454,df=8)
[1] 0.7613218
> 1-pt(2.236,df=8)
[1] 0.02788622
> pt(2.236,df=9)
[1] 0.9739085
> pt(2.236,df=8)
[1] 0.9721138
> qt(.95,df=8)
[1] 1.859548
> qt(.975,df=8)
[1] 2.306004
> 0.0589^3
[1] 0.0002043365
```